

CSE260 Assignment - 1

Deadline - 28th October, 11:59 PM

Assignment must be handwritten. Scan and upload PDF in the given google form

1. Convert the following decimal number to equivalent binary numbers: [5]
(a) $(4195.25)_{10}$
(b) $(2356.54)_{10}$
[for infinite fractional part, just do 4 steps and use dots for the rest]
2. Perform the following base conversions [2x4 = 8]
(a) $(A9)_{11} = (?)_7$
(b) $(0011)_{BCD} = (?)_5$
(c) $(1036)_{10} = (?)_{\text{Excess3}}$
(d) $(1110010011.10101000101011)_2 = (?)_{16}$
3. Perform **addition**, **subtraction**, and **multiplication** for the pair of following base-16 numbers. Verify your results by converting the problem into decimal. [10]
 $(A3)_{16}$
 $(47)_{16}$
4. Subtract $(499)_{10}$ from $(91)_{10}$ in 10 bits using 2's complement number system and justify whether there is an overflow or not. [5]
5. You are part of a university research team working on a prototype for a **smart agriculture monitoring system**. Your team recently received a grant of $(4F0)_{16}$ dollars from an international tech foundation to cover component costs and team bonuses. The team is buying $(26)_8$ temperature sensors, each costing $(47)_8$ dollars. You also spent $(101110010)_2$ dollars on soil moisture sensors. Once these purchases were made, your lead engineer decided to **divide the remaining money equally** among the $(C)_{16}$ core engineers as a performance bonus.
 - a. Find the total cost of all the temperature sensors, showing your workings in **Base-8**. [5]
 - b. Find the remaining money after all purchases were made, showing your workings in **hexadecimal (Base-16)**. [5]

- c. Calculate how much bonus each engineer received, showing your workings in **hexadecimal**. [5]

Ungraded Problems [NO NEED TO SUBMIT, ONLY FOR PRACTICE]

6. Which one of the following numbers is the largest?
 $(101101)_2$, $(57)_8$, $(35)_{10}$, $(1F)_{16}$
[Hint: Convert all of them to the same number system]
7. Perform **addition**, **subtraction**, and **multiplication** for the pair of following base-8 numbers. Verify your results by converting the problem into decimal.
 $(417)_8$
 $(134)_8$
8. Add $(211)_{10}$ with $(312)_{10}$ in 10 bits using 2's complement number system and justify whether there is an overflow or not.